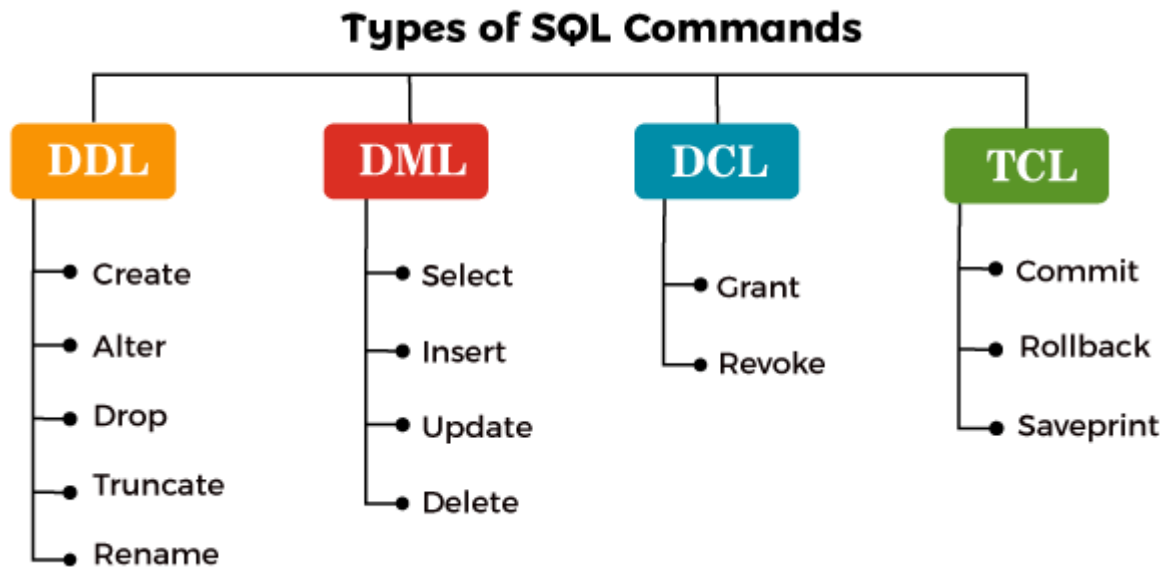


SQL Commands Classification

SQL commands are categorized into several types based on their functionality.



1. Data Definition Language (DDL)

Definition:

DDL commands are used to define or modify the structure of database objects like tables, schemas, or indexes.

Key Commands and Examples

Command	Purpose	Example
CREATE	Creates a new database object (e.g., table, database, schema, etc.).	<code>sql CREATE TABLE employees (id INT PRIMARY KEY, name VARCHAR(50), salary DECIMAL(10, 2));</code>
ALTER	Modifies an existing database object (e.g., adding a column to a table).	<code>sql ALTER TABLE employees ADD COLUMN age INT;</code>
DROP	Deletes an entire database object (e.g., table or database).	<code>sql DROP TABLE employees;</code>
TRUNCATE	Removes all rows from a table without logging individual row deletions.	<code>sql TRUNCATE TABLE employees;</code>

2. Data Manipulation Language (DML)

Definition:

DML commands are used to manipulate data stored in database tables.

Key Commands and Examples

Command	Purpose	Example
INSERT	Adds new records (rows) into a table.	sql INSERT INTO employees (id, name, salary) VALUES (1, 'Alice', 50000);
UPDATE	Modifies existing records in a table.	sql UPDATE employees SET salary = 55000 WHERE name = 'Alice';
DELETE	Removes specific records from a table.	sql DELETE FROM employees WHERE id = 1;

3. Data Query Language (DQL)

Definition:

DQL is primarily used to query data from tables.

Key Commands and Examples

Command	Purpose	Example
SELECT	Retrieves data from one or more tables.	sql SELECT name, salary FROM employees WHERE salary > 40000;

Note: Although `SELECT` is the primary DQL command, it is often grouped with DML for simplicity.

4. Data Control Language (DCL)

Definition:

DCL commands manage access rights and permissions on database objects.

Key Commands and Examples

Command	Purpose	Example
GRANT	Grants specific permissions to a user or role.	sql GRANT SELECT, INSERT ON employees TO user1;
REVOKE	Removes permissions granted to a user or role.	sql REVOKE SELECT, INSERT ON employees FROM user1;

5. Transaction Control Language (TCL)

Definition:

TCL commands manage transactions within a database to ensure data consistency

Key Commands and Examples

Command	Purpose	Example
COMMIT	Saves changes made by a transaction permanently.	sql COMMIT;
ROLLBACK	Reverts changes made by a transaction to its previous state.	sql ROLLBACK;
SAVEPOINT	Sets a point within a transaction to roll back to later.	sql SAVEPOINT sp1; UPDATE employees SET salary = 60000 WHERE id = 2; ROLLBACK TO sp1;

Summary Table

Category	Purpose	Examples
DDL	Define and modify database structures.	CREATE , ALTER , DROP , TRUNCATE
DML	Manipulate data in the tables.	INSERT , UPDATE , DELETE
DQL	Query data from the database.	SELECT
DCL	Control access and permissions.	GRANT , REVOKE
TCL	Manage transactions to ensure data integrity.	COMMIT , ROLLBACK , SAVEPOINT